

Request Processing Database ERD

PostgreSQL database: hrms_requests

requests		
Key	Column	Type
PK	id	UUID
	user_id	UUID
	request_type	VARCHAR(100)
	payload	TEXT
	status	VARCHAR(30)
	result	TEXT
	error_message	TEXT
	created_at	TIMESTAMPTZ
	updated_at	TIMESTAMPTZ
	completed_at	TIMESTAMPTZ
	version	BIGINT
	idempotency_key	VARCHAR(100)
	idempotency_fingerprint	VARCHAR(64)

Constraints

- UNIQUE (user_id, idempotency_key)
- status ∈ { PENDING, PROCESSING, COMPLETED, FAILED }
- version ≥ 0



Physical FK

processed_events		
Key	Column	Type
PK FK	event_id	UUID
	request_id	UUID
	event_type	VARCHAR(100)
	event_version	INTEGER
	occurred_at	TIMESTAMPTZ
	processed_at	TIMESTAMPTZ

Constraints

- FOREIGN KEY (request_id) REFERENCES requests(id) ON DELETE CASCADE
- event_version > 0



Logical Reference
(No Physical FK)

outbox_events		
Key	Column	Type
PK	id	UUID
	aggregate_id	UUID
	aggregate_type	VARCHAR(100)
	event_type	VARCHAR(100)
	payload	JSONB
	status	VARCHAR(30)
	retry_count	INTEGER
	created_at	TIMESTAMPTZ
	published_at	TIMESTAMPTZ
	last_error	TEXT

Constraints

- status ∈ { PENDING, PROCESSING, PUBLISHED, FAILED }
- retry_count ≥ 0

Legend



= Physical Foreign Key



= Logical Reference (No Physical Foreign Key)

PK = Primary Key

FK = Foreign Key

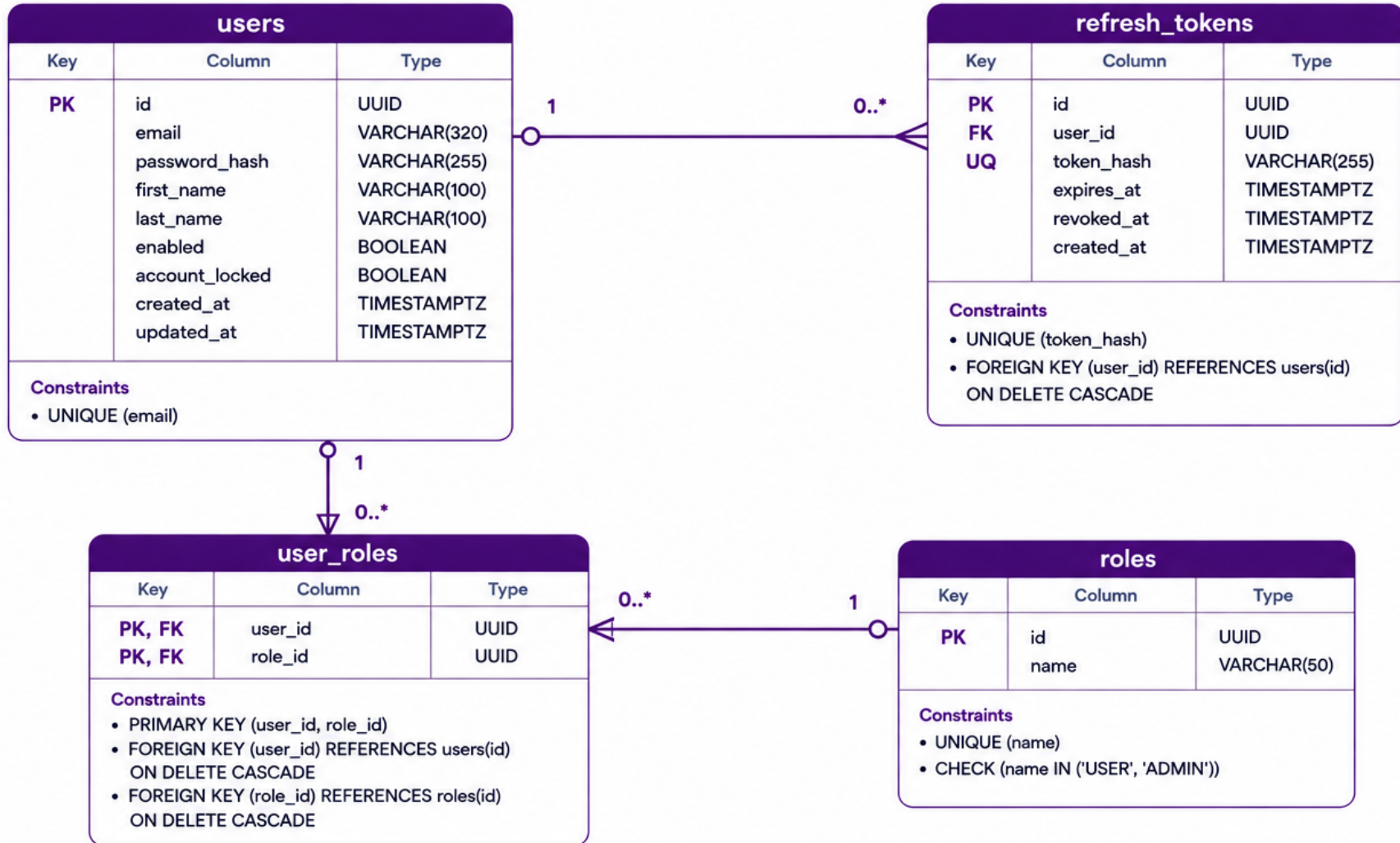
UQ = Unique Constraint

Notes

- requests.user_id is a logical identity from Auth domain (no physical foreign key).
- outbox_events.aggregate_id logically identifies the request aggregate.
- processed_events records processed event IDs to prevent duplicate event processing.
- All timestamp columns use PostgreSQL TIMESTAMPTZ.

Authentication Database ERD

PostgreSQL database: hlrms_auth



Legend

○ —> = Physical Foreign Key (with ON DELETE CASCADE)

PK = Primary Key FK = Foreign Key

UQ = Unique Constraint

Notes

- password_hash and token_hash store hashed credential/token values.
- refresh_tokens stores hashed refresh tokens and supports token revocation.
- user_roles implements a many-to-many relationship between users and roles.
- All timestamp columns use PostgreSQL TIMESTAMPTZ.

Transfer Demo Database ERD

PostgreSQL database: hlrms_transfer

transfer_profiles		
Key	Column	Type
PK UQ UQ	id	UUID
	user_id	UUID
	transfer_id	VARCHAR(16)
	email	VARCHAR(320)
	display_name	VARCHAR(200)
	created_at	TIMESTAMPTZ
	updated_at	TIMESTAMPTZ
Constraints • UNIQUE (user_id) • UNIQUE (transfer_id)		

Participant Identity References
sender_user_id / recipient_user_id
(No Physical FK)

wallet_balances		
Key	Column	Type
PK FK	id	BIGSERIAL
	user_id	UUID
	currency	VARCHAR(3)
	balance	NUMERIC(20, 2)
	version	BIGINT
	created_at	TIMESTAMPTZ
	updated_at	TIMESTAMPTZ
Constraints • FOREIGN KEY (user_id) REFERENCES transfer_profiles(user_id) ON DELETE CASCADE • UNIQUE (user_id, currency) • CHECK (balance >= 0) • CHECK (currency IN ('USD', 'EUR', 'TRY', 'SYP'))		

transfer_transactions		
Key	Column	Type
PK UQ	id	UUID
	request_id	UUID
	sender_user_id	UUID
	recipient_user_id	UUID
	sender_transfer_id	VARCHAR(16)
	recipient_transfer_id	VARCHAR(16)
	sender_display_name	VARCHAR(200)
	recipient_display_name	VARCHAR(200)
	amount	NUMERIC(20, 2)
	currency	VARCHAR(3)
	created_at	TIMESTAMPTZ
	Constraints • UNIQUE (request_id) • CHECK (amount > 0) • CHECK (sender_user_id <> recipient_user_id)	

Legend

— Physical Foreign Key

- - - Logical Reference (No Physical FK)

PK = Primary Key

FK = Foreign Key

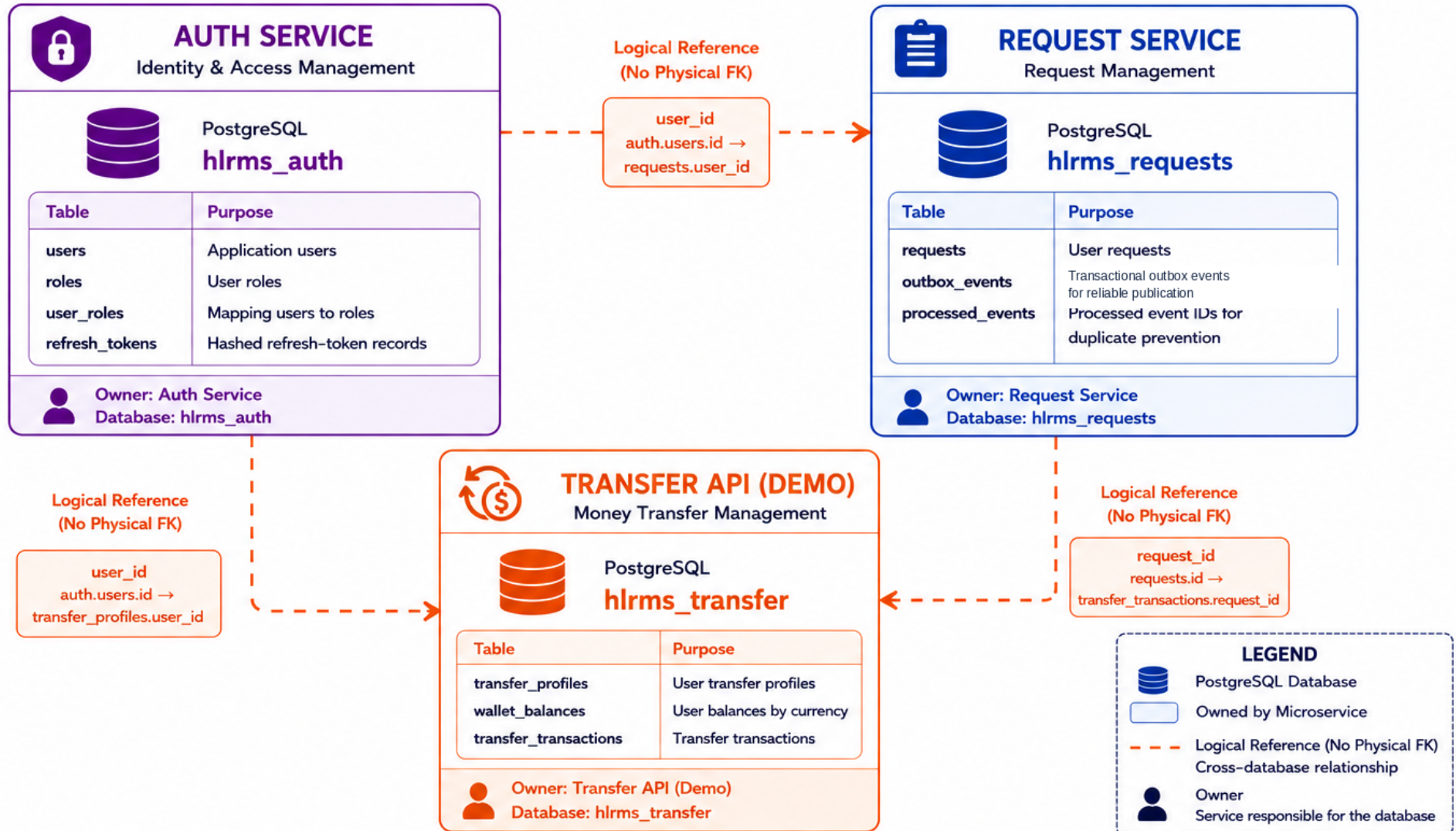
UQ = Unique Constraint

Notes

- transfer_profiles.user_id is a logical identity from the Auth domain.
- transfer_transactions.request_id logically references HLRMS requests.id.
- sender_user_id and recipient_user_id are logical participant identities; no database FK is declared.
- All timestamp columns use PostgreSQL TIMESTAMPTZ.

HLRMS Database Landscape

Logical Data Ownership & Cross-Database References



NOTES

- Each application domain owns and manages its database boundary.
- No cross-database foreign keys are created.
- Relationships between databases are logical references only.
- Auth Service is the identity source for users across the system.
- Request Service uses user_id to identify who created the request.
- Transfer API uses user_id for wallet/profile ownership and request_id for traceability.